

DBMS — Lecture 5

Integrity Constraints

Learning Objectives

Understand integrity constraints.

Differentiate domain, key, entity integrity, and referential integrity.

Understand why constraints are important.

Integrity Constraints

Integrity constraints are rules that ensure database data remains accurate, valid, and consistent.

Major Constraints

Domain Constraint: Attribute values must come from an allowed domain. Example: Age must be an integer within an appropriate range.

Key Constraint: Key values must uniquely identify tuples.

Entity Integrity: A primary key cannot be NULL.

Referential Integrity: A foreign key must match a referenced primary-key value or be NULL when permitted.

NOT NULL: Prevents NULL values in an attribute.

UNIQUE: Prevents duplicate values in a column or column combination.

CHECK: Enforces a condition on values.

Example

`Student(Student_ID PRIMARY KEY, Name NOT NULL, Age CHECK(Age >= 17))`

If Enrollment.Student_ID is a foreign key referencing Student.Student_ID, it must refer to an existing student (subject to the allowed NULL rule).

Why Important?

Prevents invalid records.

Maintains consistency between related tables.

Reduces application-level errors.

Improves reliability and data quality.

Key Takeaway

Integrity constraints enforce rules that keep database data correct and consistent.